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TUTORIAL

# Feedback

*Negative Feedback in Analog CMOS Amplifiers: Properties, Topologies, and Loading Effects*

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

**Based on Design of Analog CMOS Integrated Circuits, Chapter 8 (Razavi)**

OUTLINE

# Roadmap for This Lecture

*Six sections, from feedback fundamentals to loading effects*

## I General Consideration of Feedback

- Closed-loop transfer function Y/X = H/(1+GH)
- Loop gain and the deep-feedback intuition

## II Properties of Feedback

- Gain desensitization
- Bandwidth modification and the gain-bandwidth trade-off

## III Amplifier Types and Sensing/Return Mechanisms

- Four ideal amplifier types: voltage, transimpedance, transconductance, current
- Sensing (shunt) vs. return (series) mechanisms

## IV Feedback Topologies I — Voltage-Voltage & Current-Voltage

- Series-shunt: voltage amplifier, Rin high, Rout low
- Series-series: transconductance amplifier, Rin high, Rout high

## V Feedback Topologies II — Voltage-Current & Current-Current

- Shunt-shunt: transimpedance amplifier and the optical-receiver application
- Shunt-series: current amplifier

## VI Effect of Loading of the Feedback Network

- Two-port network models: Z, Y, H, G parameters
- Computing the loaded open-loop gain

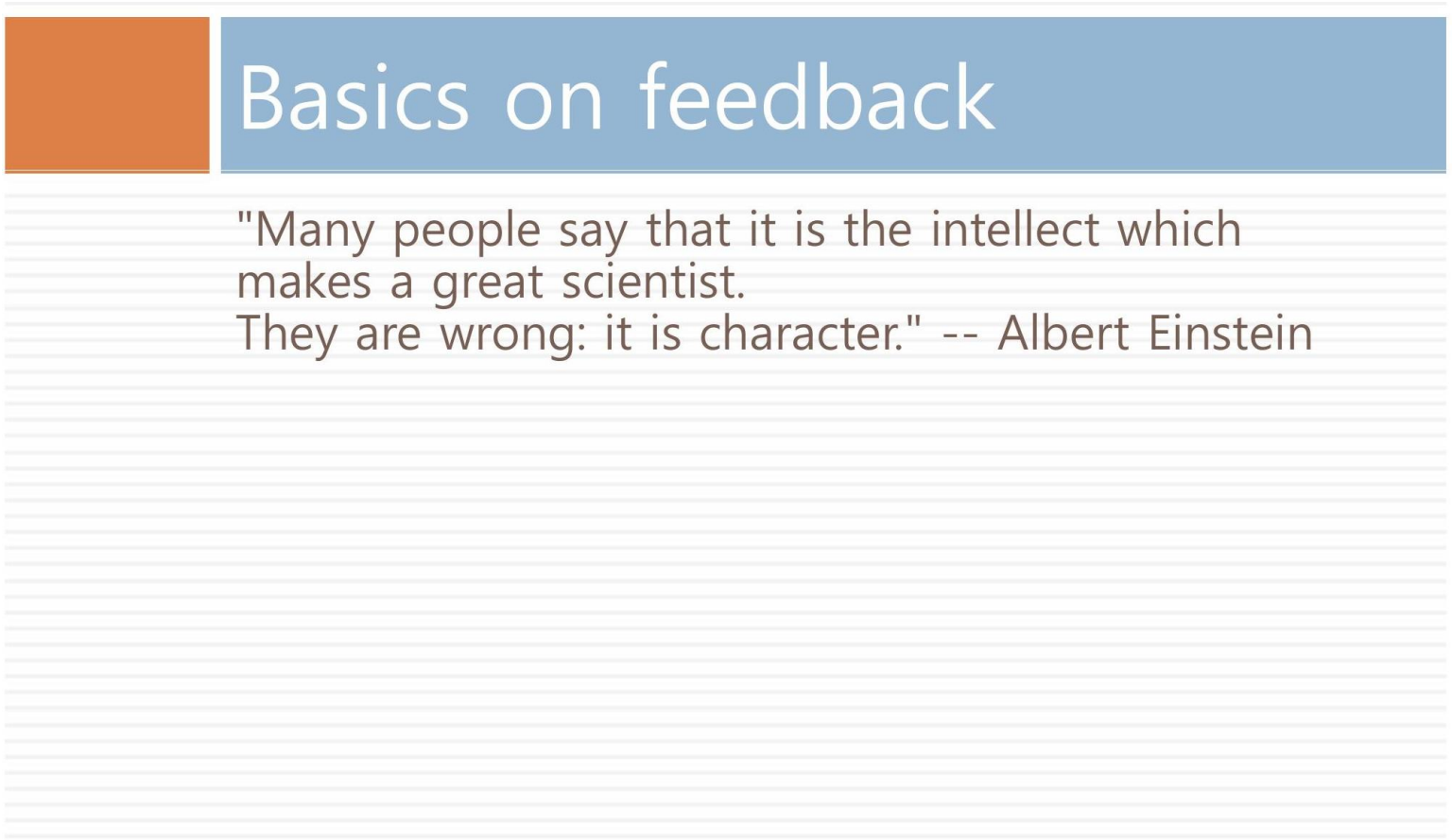
I

# General Consideration of Feedback

- Closed-loop transfer function $Y/X = H/(1+GH)$
- Loop gain and the deep-feedback intuition

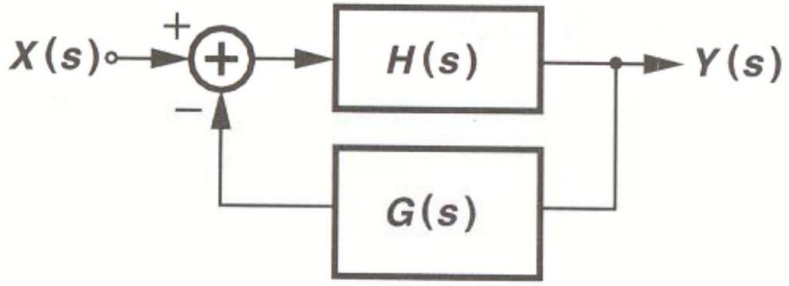
# Opening: The Spirit of Feedback Design

- Einstein quote framing the chapter; previews the roadmap: general consideration, key properties, four feedback topologies, and effect of loading
- Chapter 8 previews: gain desensitization, bandwidth extension, terminal impedance modification, and improved linearity from negative feedback



# General Block Diagram and Closed-Loop Transfer Function

- Summing node forms the error signal X(s) − G(s)Y(s), driven through the feedforward block H(s) to produce Y(s)
- Solving Y(s) = H(s)[X(s) − G(s)Y(s)] gives Y(s)/X(s) = H(s)/[1+G(s)H(s)]
- G(s)H(s) is the loop gain (loop transmission); the factor (1+loop gain) governs every benefit feedback provides

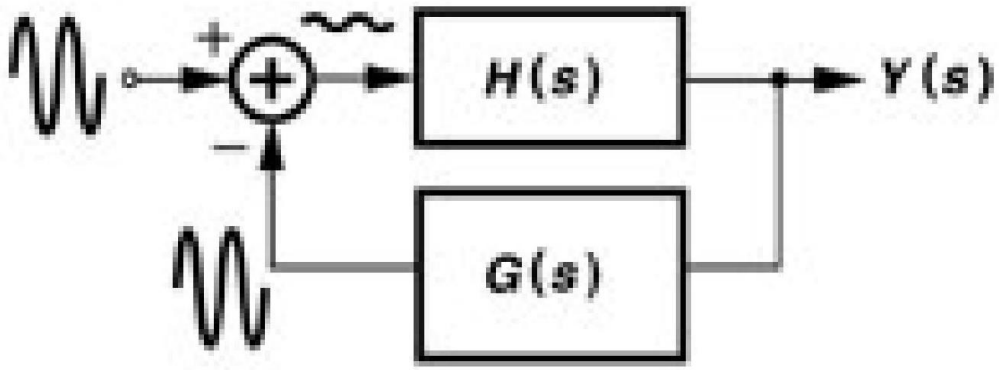



$$Y(s) = H(s)[X(s) - G(s)Y(s)]$$

$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

# Intuition for Deep Feedback: G(s)Y(s) Tracks X(s)

- Sinusoidal illustration shows the feedback signal G(s)Y(s) nearly reproduces the input X(s) when the loop gain is large
- For large loop gain, Y(s) ≈ X(s)/G(s) — the response depends mainly on the accurate, passive feedback network G(s), not the imprecise amplifier H(s)




Similarity between output of feedback network and input signal

II

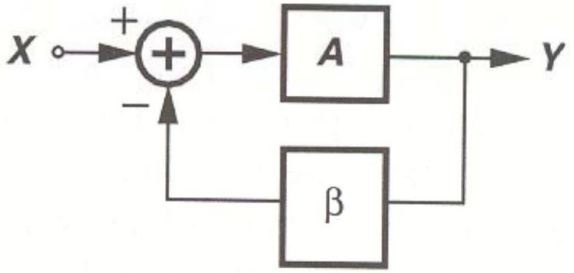
# Properties of Feedback

- Gain desensitization
- Bandwidth modification and the gain-bandwidth trade-off
- Terminal impedance modification

# Gain Desensitization — General Derivation

- Closed-loop gain A_CL = A/(1+Aβ) = (1/β)·[Aβ/(1+Aβ)]; for large loop gain Aβ, A_CL ≈ 1/β, independent of the imprecise open-loop gain A
- Alternative form A_CL = A_CLI·(−T)/(1−T) uses the return-ratio convention T = −Aβ, with A_CLI = 1/β the ideal closed-loop gain as loop gain → ∞

Gain desensitization

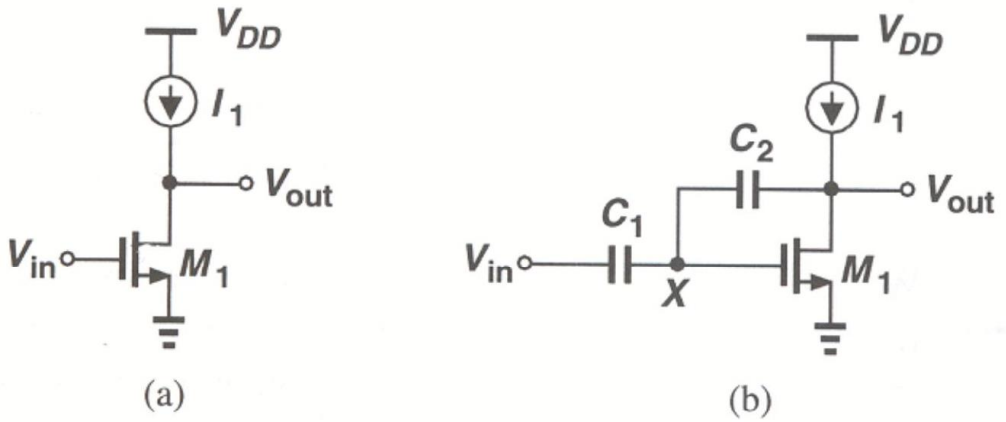



$$A_{CL} = \frac{Y}{X} = \frac{A}{1+A\beta} = \frac{1}{\beta}\cdot\frac{A\beta}{1+A\beta} = A_{CLI}\,\frac{-T}{1-T}$$

# Gain Desensitization — Capacitive-Divider Feedback Circuit

- (a) Open-loop common-source stage: M1 with ideal current-source load I1, gain ≈ −gm1·ro1
- (b) Adds a capacitive divider C1 (Vin to node X) and C2 (X to Vout) as the feedback network, β ≈ C2/(C1+C2); node X is a floating high-impedance node at the MOSFET gate

Gain desensitization



# Gain Desensitization — KCL Proof: Vout/Vin → −C1/C2

- Displacement-current balance at node X: (Vout−VX)C2s = (VX−Vin)C1s, combined with VX = −Vout/(gm1ro1)
- For large intrinsic gain gm1ro1, Vout/Vin → −C1/C2 — closed-loop gain set purely by a precisely matched capacitor ratio

$$(V_{OUT} - V_X)C_2 s = (V_X - V_{in})C_1 s$$

$$V_X = -V_{OUT} / g_{m1} r_{o1}$$

$$\Rightarrow \frac{V_{OUT}}{V_{IN}} = -\frac{1}{\left(1 + \frac{1}{g_{m1} r_{o1}}\right)\frac{C_2}{C_1} + \frac{1}{g_{m1} r_{o1}}}$$

$$= -\frac{C_1}{C_2}, \text{ for large } g_{m1} r_{o1}$$

# Bandwidth Modification — Algebraic Derivation

- Single-pole open-loop amplifier A = $A_0/(1+jf/fp)$ substituted into $A_CL = (1/\beta)[A\beta/(1+A\beta)]$
- Result: $A_CL = (1/\beta)\cdot\{A_0\beta/[(1+A_0\beta)(1+j(f/fp)(1+A_0\beta))]\}$ — DC gain drops to $\approx 1/\beta$ while the −3dB frequency extends to $fp(1+A_0\beta)$

**Bandwidth Modification**

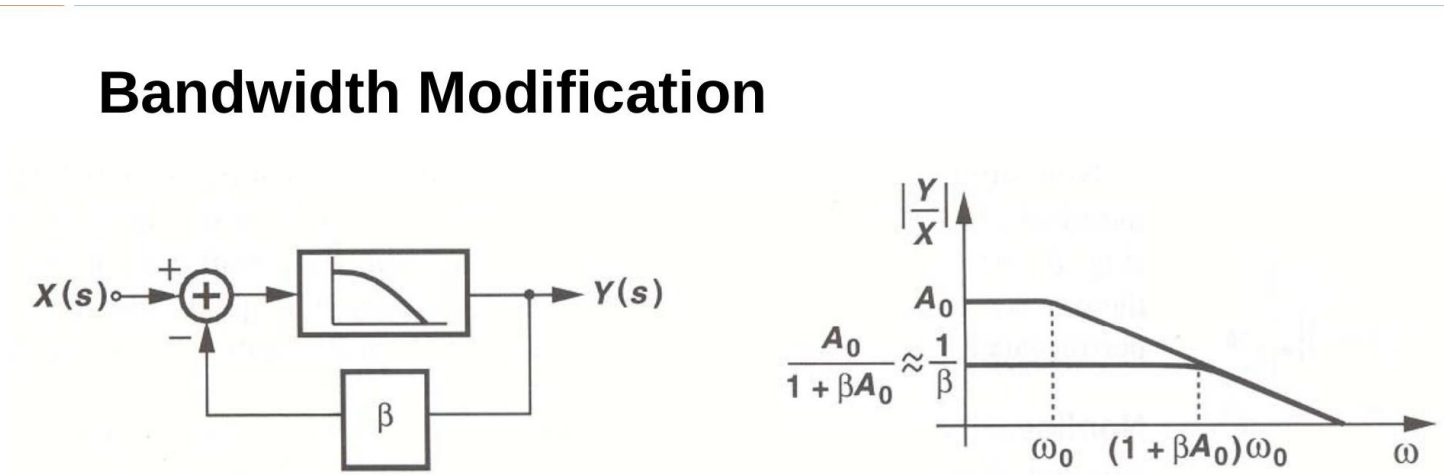
$$Given \quad A = \frac{A_0}{\left(1 + j\,f/f_p\right)}$$

$$A_{CL} = \frac{1}{\beta}\frac{A\beta}{1+A\beta} = \frac{1}{\beta}\frac{\dfrac{A_0\beta}{\left(1+j\,f/f_p\right)}}{1+\dfrac{A_0\beta}{\left(1+j\,f/f_p\right)}} = \frac{1}{\beta}\frac{A_0\beta}{1+j\,f/f_p + A_0\beta}$$

$$= \frac{1}{\beta}\frac{A_0\beta}{(1+A_0\beta)\left(1+j\,f/f_p(1+A_0\beta)\right)}$$

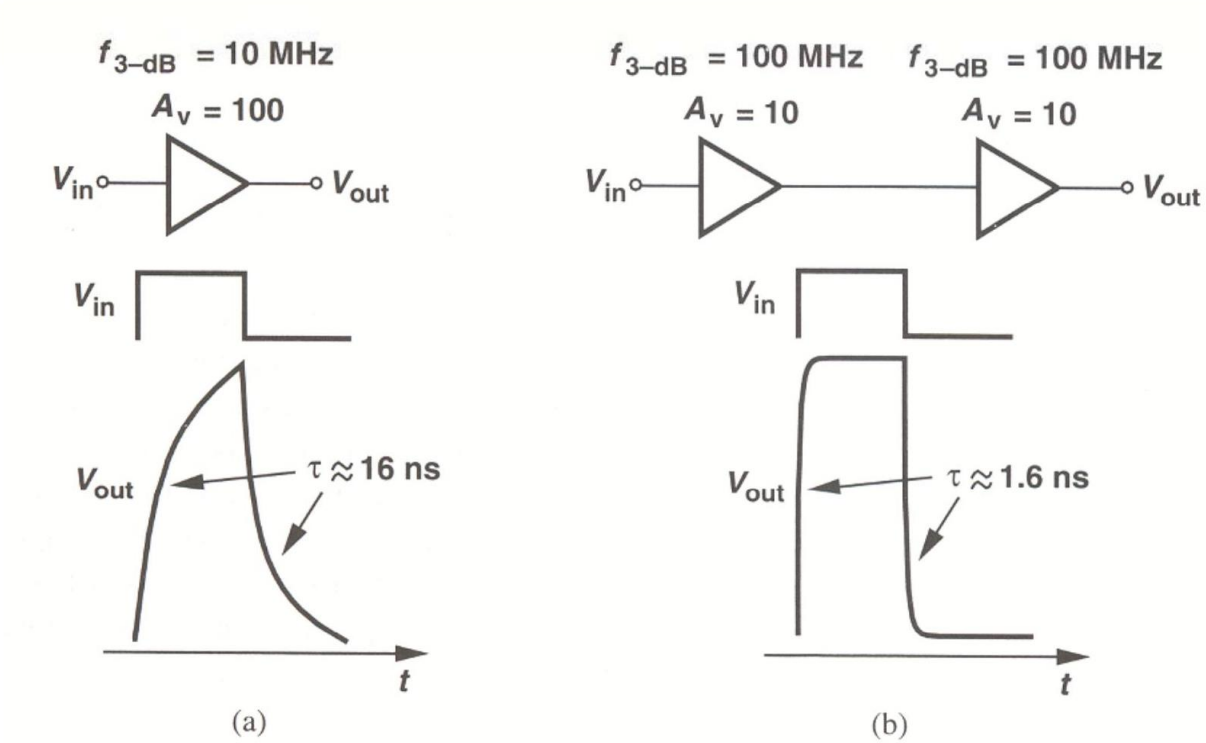
# Bandwidth Modification — Bode-Plot Interpretation

- Open-loop response is flat at $A_0$ up to $\omega_0$ then rolls off; closed-loop response is flat at a lower level $\approx 1/\beta$ but extends to $(1+\beta A_0)\omega_0$ before rolling off
- At high frequency both curves converge to the same slope since loop gain has decayed — illustrates the constant gain-bandwidth product



# Bandwidth Trade-off Example — Amplifying a 20-MHz Square Wave

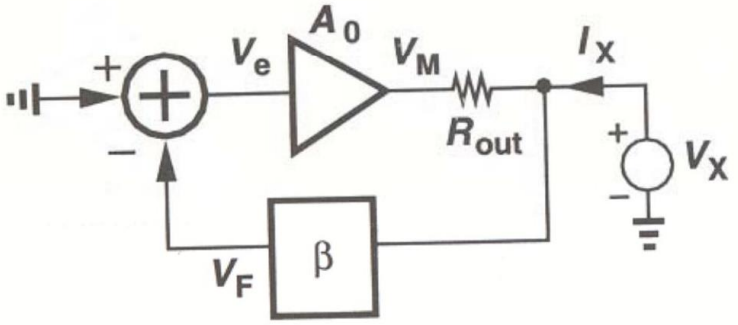
- (a) Single stage, Av=100, f3dB=10MHz (below the 20MHz fundamental): output badly distorted, exponential edges with τ≈16ns
- (b) Cascade of two feedback stages, Av=10 each (same total Av=100) but f3dB=100MHz each: sharp edges, τ≈1.6ns — favors multiple lower-gain, wider-band feedback stages for wideband signals



# Terminal Impedance Modification — Output Resistance (General)

- Test source VX/IX applied at the output with the summing input set to zero; VM = A₀Ve = −βA₀VX since Ve = −VF = −βVX
- IX = VX(1+βA₀)/Rout ⇒ Rout,CL = Rout/(1+βA₀) — voltage sensing (shunt at the output) always lowers output resistance by the loop-gain factor

**Terminal impedance modification**

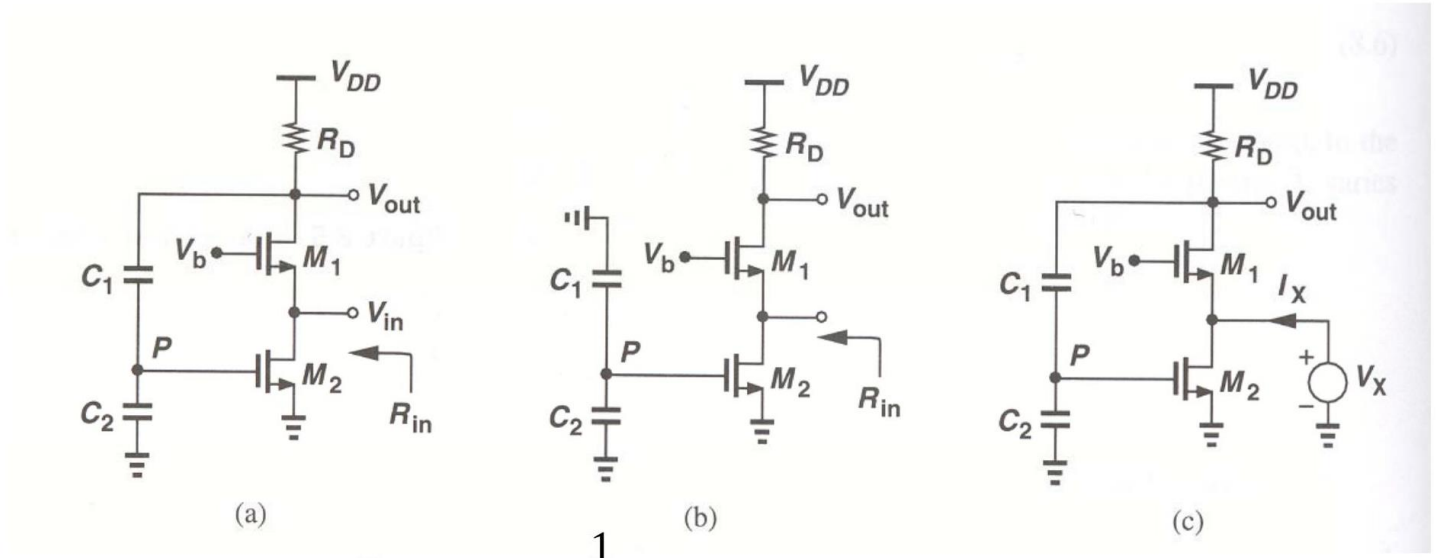



$$I_X = \frac{V_X - V_M}{R_{out}} = \frac{V_X - (-\beta A_0 V_X)}{R_{out}}$$

$$\frac{V_X}{I_X} = R_{out,CL} = \frac{R_{out}}{1+\beta A_0}$$

# Local Feedback Example — Input Resistance of a Cascode Node

- Cascode M1 (gate at fixed Vb) over M2 (gate at node P); a capacitive divider C1 (Vout to P), C2 (P to ground) feeds a local feedback loop into M2's gate
- Open-loop: Rin,open = 1/(gm1+gmb1); closed-loop: Rin,closed = Rin,open/(1+A_OLβ), with A_OLβ = [C1/(C1+C2)]·gm2·RD



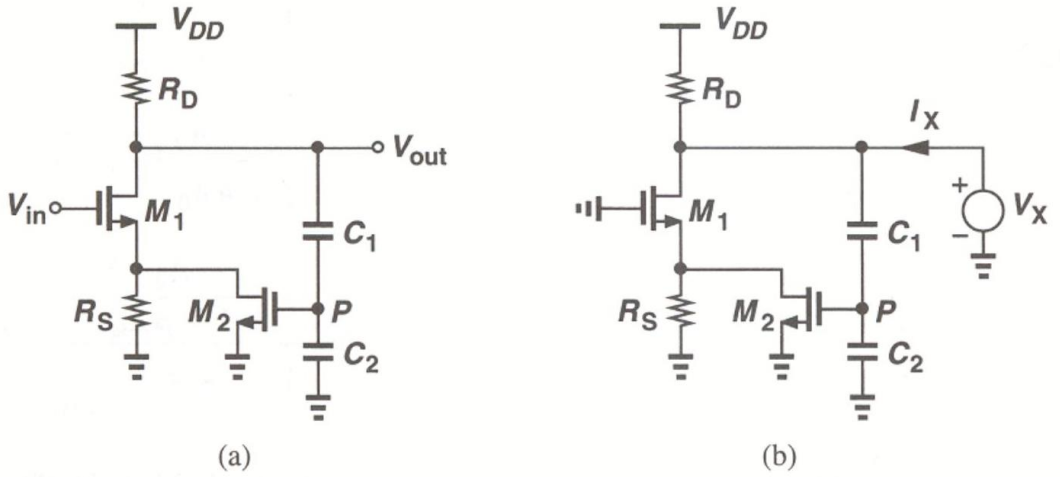

$$R_{in,open} = \frac{1}{g_{m1} + g_{mb1}}$$

$$R_{in,closed} = R_{in,open}/(1 + A_{OL}\beta)$$

$$A_{OL}\beta = \frac{C_1}{C_1 + C_2} g_{m2} R_D$$

# Local Feedback Example — Output Resistance with Source Degeneration

- M1 (common-source, source-degenerated by RS) with M2 sensing Vout via a C1/C2 divider and injecting current back into M1's source node
- Rout,closed = Rout,open/(1+A_OLβ), with A_OLβ = [C1/(C1+C2)]·gm2·(RS∥1/(gm1+gmb1))·(gm1+gmb1)RD — three cascaded stage gains multiplied together




$$R_{out,open} = R_D$$

$$R_{out,closed} = R_{out,open}/(1+A_{OL}\beta)$$

$$A_{OL}\beta = \left\{\frac{C_1}{C_1+C_2}\right\}\left\{g_{m2}\left(R_S \parallel \frac{1}{g_{m1}+g_{mb1}}\right)\right\}\left\{(g_{m1}+g_{mb1})R_D\right\}$$

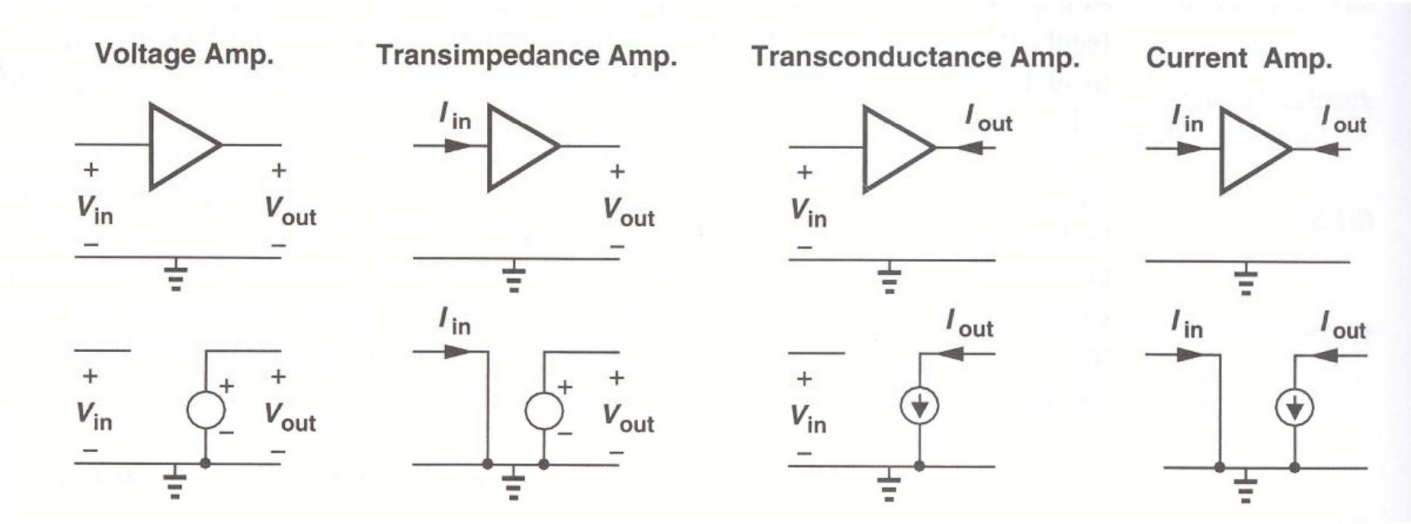
III

# Amplifier Types and Sensing/Return Mechanisms

- Four ideal amplifier types: voltage, transimpedance, transconductance, current
- Sensing (shunt) vs. return (series) mechanisms
- Practical sensing/summing circuits

# Four Ideal Amplifier Types

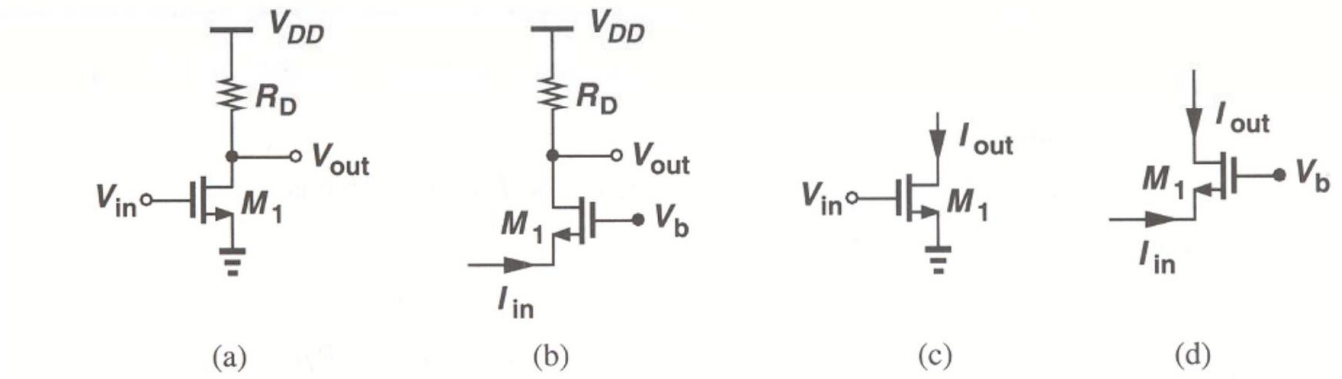
- Voltage Amp (Vin→Vout): Rin=∞, Rout=0; Transimpedance Amp (Iin→Vout): Rin=0, Rout=0
- Transconductance Amp (Vin→Iout): Rin=∞, Rout=∞; Current Amp (Iin→Iout): Rin=0, Rout=∞
- These four ideal types map directly onto the four feedback topologies covered next: series-shunt, shunt-shunt, series-series, shunt-series



# Single-Transistor Implementations of the Four Amplifier Types

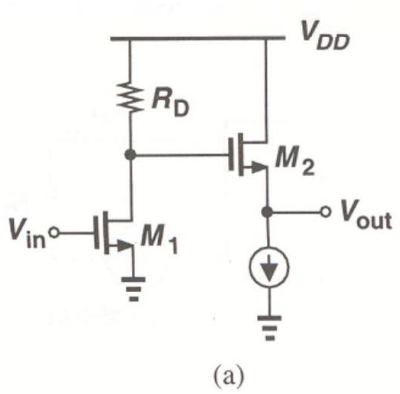
- (a) Voltage amp: common-source stage with RD load; (b) transimpedance amp: common-gate stage, Iin into the source, Vout at the drain
- (c) Transconductance amp: common-source with no drain resistor, Iout taken directly from the drain; (d) current amp: common-gate stage, Iout taken directly from the drain

Simple implementations of four types of amplifiers

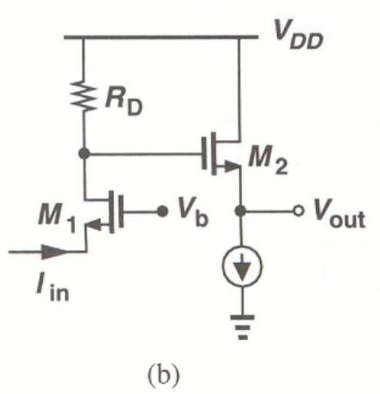


# Two-Transistor Amplifiers with Improved Rout and Current Gain

- "Small output resistance" (a, b): adding a source-follower-like M2 stage after M1 lowers the output impedance seen at Vout
- "Large current gain" (c, d): cascading M1 and M2 gives Iout = −gm2gm1RD·Vin (voltage input) or Iout = gm2RD·Iin (current input)

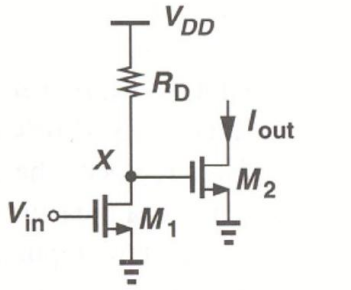



(a)

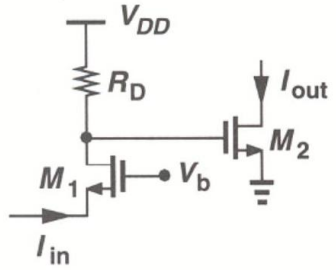



(b)

**small output resistance**




$$I_{out} = -g_{m2}g_{m1}R_D V_{in}$$

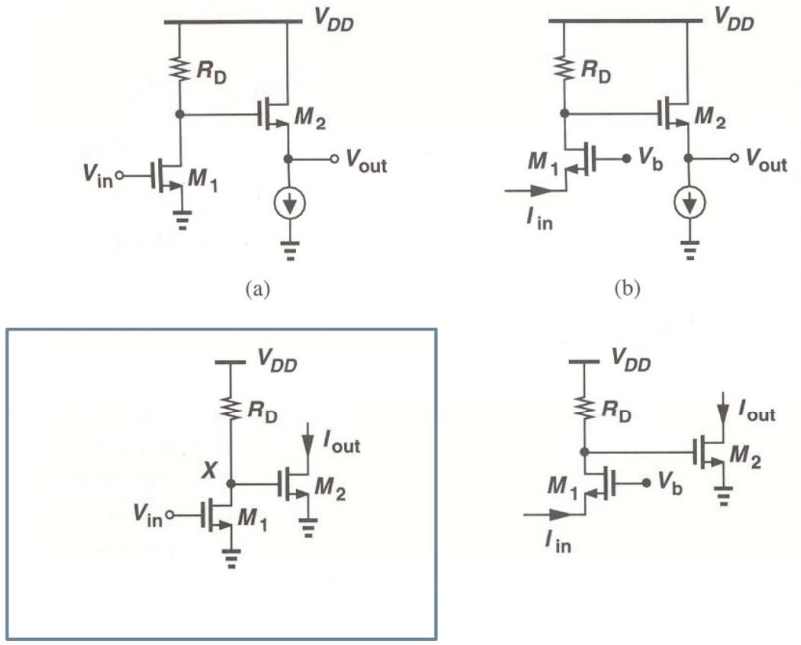



$$I_{out} = g_{m2}R_D I_{in}$$

**large current gain**

# Example 8.1 — Transconductance Gain of the Cascaded Stage

- Circuit: M1 (Vin at gate, drain at node X via RD∥ro1) drives M2 (gate at X, drain delivers Iout with no load resistor)
- Gm = −gm1(ro1∥RD)·gm2 — the product of stage 1's voltage gain and stage 2's transconductance

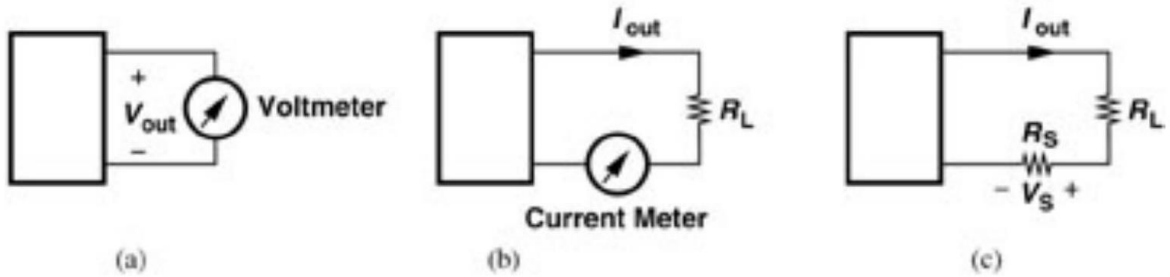



Calculate the transconductance gain

$$G_m = -g_{m1}(r_{o1} \parallel R_D)g_{m2}$$

# Sensing Mechanisms: Shunt (Voltmeter) vs. Series (Ammeter)

- (a) Voltage sensing: voltmeter placed in parallel (shunt) with the output port; (b) current sensing: current meter placed in series with RL
- (c) A small series sense resistor RS measures Iout via its voltage drop — a practical stand-in for an ideal ammeter
- Naming convention: shunt sensing/return uses a voltmeter-like (parallel) connection; series sensing/return uses an ammeter-like (series) connection



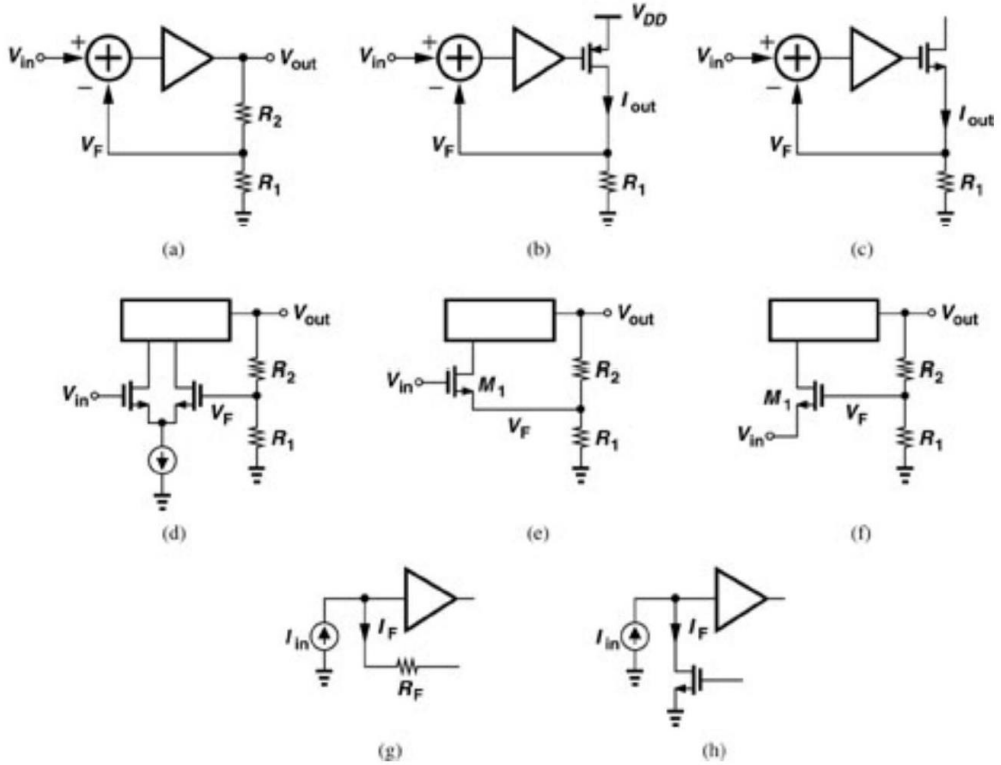

Sensing (a) a voltage by a voltmeter, (b) a current by a current meter, (c ) a current by a small resistor

Voltmeter placed in parallel with the corresponding port (shunt feedback)
Current meter placed in series with the corresponding port (series feedback)

# Practical Circuits for Sensing and Summing Voltages and Currents

- (a)–(c): differential-input triangle amplifier with an R1/R2 divider sensing Vout to generate VF — the classic non-inverting voltage-feedback realization
- (d)–(f): the same idea implemented with an actual MOS differential pair instead of an ideal triangle amplifier
- (g)–(h): current-feedback realizations — IF summed with Iin at a node, generated via RF or a voltage-to-current MOS branch

Practical means of sensing and adding voltages and currents



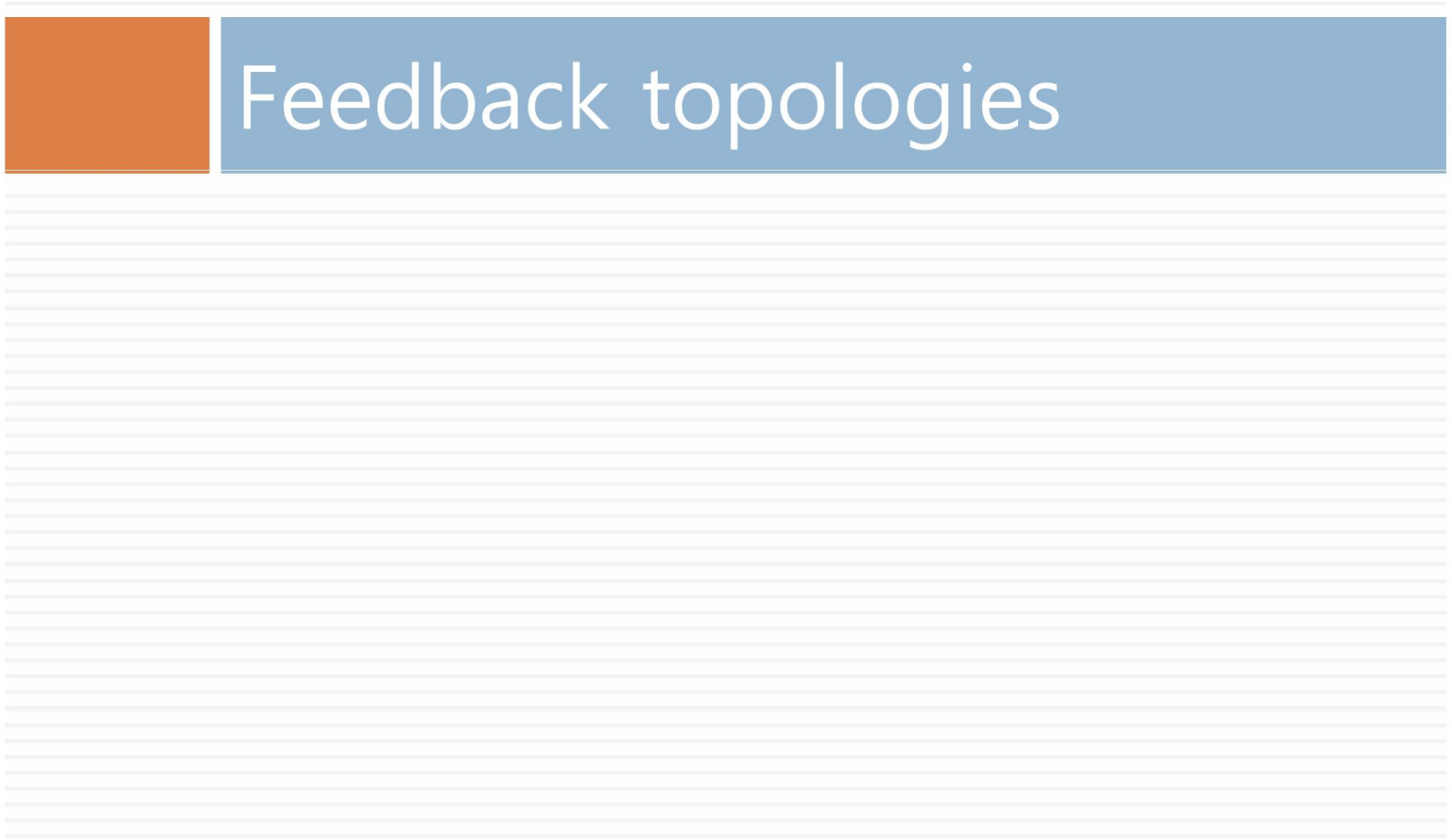
IV

# Feedback Topologies I — Voltage-Voltage & Current-Voltage

- Series-shunt: voltage amplifier, Rin high, Rout low
- Series-series: transconductance amplifier, Rin high, Rout high
- Worked examples Ex8.2–Ex8.4

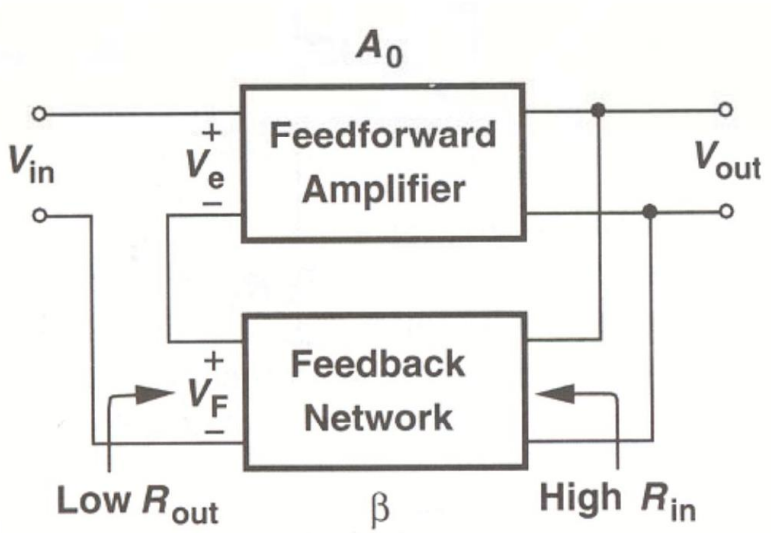
# Feedback Topologies: Four Basic Configurations

- Combining output sensing (voltage=shunt/voltmeter, current=series/ammeter) with input return (series-voltage or shunt-current) yields four topologies
- Voltage-Voltage (Series-Shunt), Current-Voltage (Series-Series), Voltage-Current (Shunt-Shunt), Current-Current (Shunt-Series)
- Each is analyzed for closed-loop gain and Rin/Rout, with worked CMOS examples Ex8.2 through Ex8.10



# Series-Shunt (Voltage-Voltage) Feedback — Block Diagram

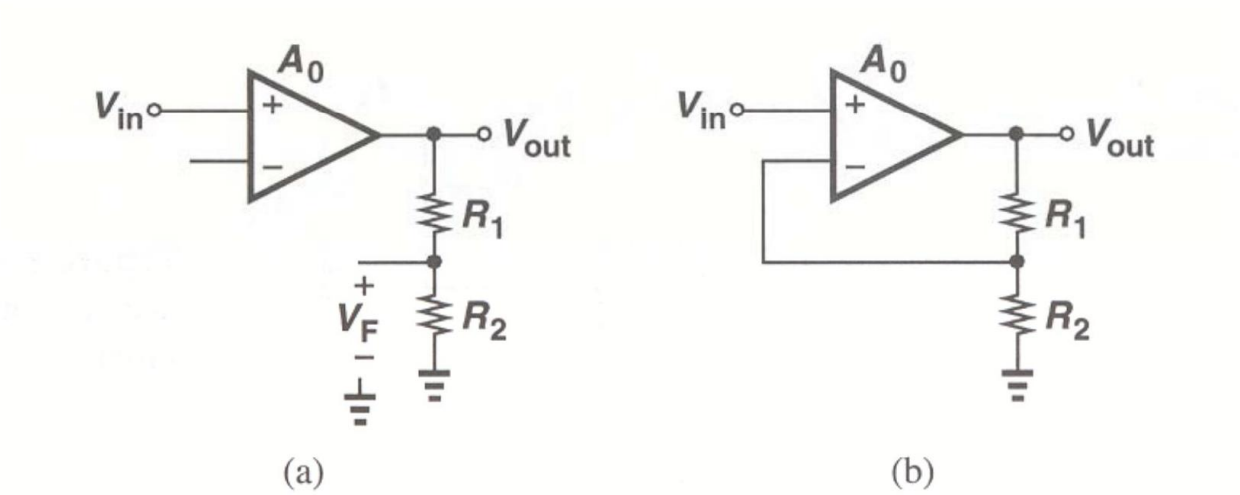
- Feedforward amplifier $A_0$ takes Ve = Vin−VF (series summing at the input); feedback network β senses Vout in shunt to produce VF
- Ideal port impedances of the feedback network itself: Low Rout at the series return port, High Rin at the shunt sensing port
- Closed-loop gain Vout/Vin = $A_0/(1+A_0\beta)$ — the standard feedback form with $A=A_0$




$$\frac{V_{out}}{V_{in}} = \frac{A_o}{1+A_o\beta}$$

# Series-Shunt Example — Resistive-Divider Voltage Sensing

- (a) Differential-input triangle amplifier $A_0$ with Vin at (+) and VF from an R1/R2 divider ($\beta = R2/(R1+R2)$) at (−)
- (b) Same circuit redrawn with the divider tap wired directly to the (−) input — the classic non-inverting voltage-voltage feedback amplifier

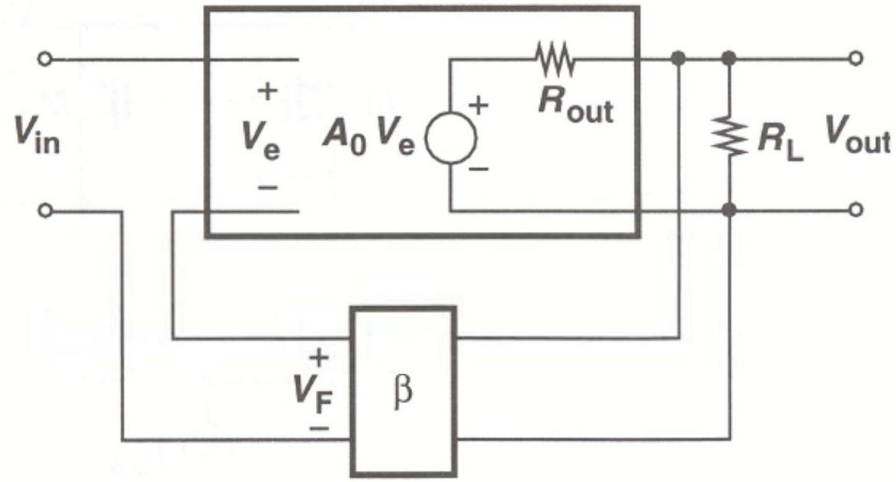



Amplifier with output sensed by a resistive divider

Voltage-voltage feedback amplifier

# Series-Shunt with Finite Rout and Load RL

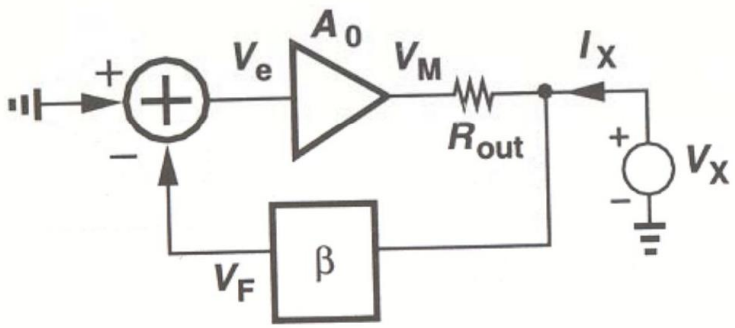
- Thévenin model adds the amplifier's output resistance Rout in series with $A_0 V_e$, feeding RL and the sensing network β
- Physical intuition: feedback forces Vout to stay nearly constant despite RL changes ⇒ effective Rout becomes smaller




The feedback amplifier is working to maintain Vout constant
=> Rout becomes smaller

# Series-Shunt — Quantitative Closed-Loop Output Resistance

- Test source at output: IX = [VX−(−βA₀VX)]/Rout = VX(1+βA₀)/Rout
- Rout,CL = Rout/(1+βA₀) — confirms voltage sensing lowers Rout by the loop-gain factor, matching the general result from Slide 11

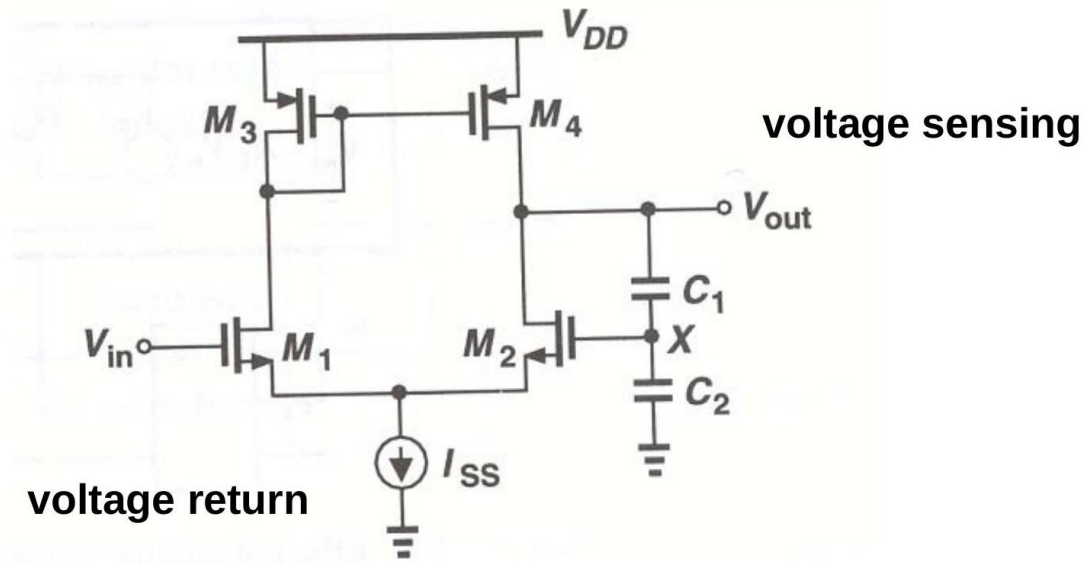



$$I_X = \frac{V_X - V_M}{R_{out}} = \frac{V_X - (-\beta A_0 V_X)}{R_{out}}$$

$$\frac{V_X}{I_X} = R_{out,CL} = \frac{R_{out}}{1+\beta A_0}$$

# Example 8.2 — Differential-Pair Realization of Series-Shunt

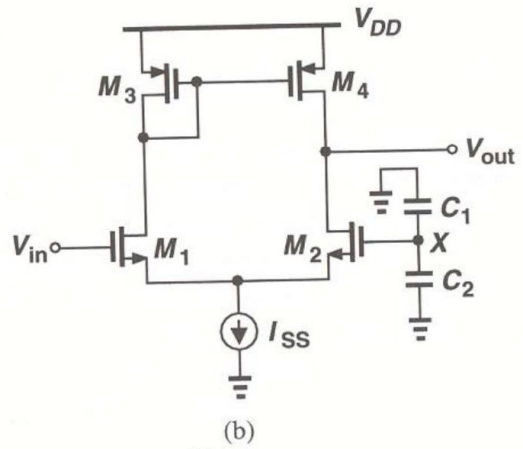
- M1/M2 differential pair with tail current ISS, PMOS active loads M3/M4; Vin drives M1's gate, M2's gate is driven by node X of a C1/C2 divider off Vout
- C1/C2 divider realizes β = C1/(C1+C2): "voltage sensing" at Vout, "voltage return" into M2's gate via the differential pair's inherent subtraction
- Goal: find the closed-loop gain and output resistance of this amplifier



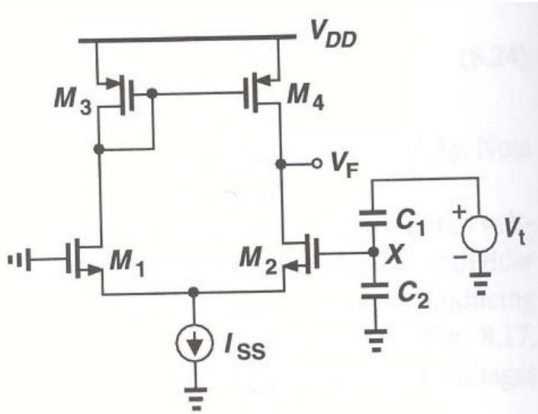

Calculate the closed-loop gain and output resistance of the amplifier

# Example 8.2 — Loop Gain by Breaking the Loop

- Break the loop at node X, inject test voltage Vt with Vin grounded; signal propagates through the pair's gain −gm1(ro2∥ro4) to Vout, then back through the C1/C2 divider to VF
- Loop gain $\beta A_0$ = [C1/(C1+C2)]·gm1(ro2∥ro4); closed-loop Rout,closed = (ro2∥ro4)/(1+$\beta A_0$)



(b)



(c)

$$V_F = -V_t \frac{C_1}{C_1 + C_2} g_{m1}(r_{o2} \| r_{o4})$$

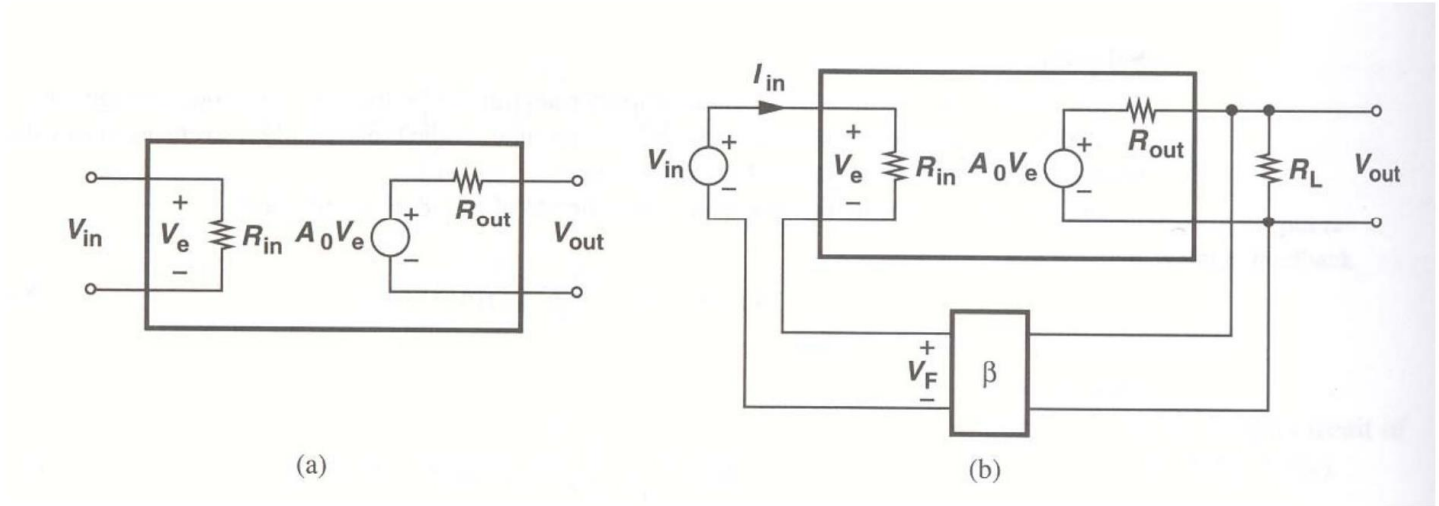
$$\beta A_o = \frac{C_1}{C_1 + C_2} g_{m1}(r_{o2} \| r_{o4})$$

$$A_{closed} = \frac{g_{m1}(r_{o2} \| r_{o4})}{1 + \frac{C_1}{C_1 + C_2} g_{m1}(r_{o2} \| r_{o4})}$$

$$R_{out,closed} = \frac{r_{o2} \| r_{o4}}{1 + \frac{C_1}{C_1 + C_2} g_{m1}(r_{o2} \| r_{o4})}$$

# Series-Shunt Input Resistance — Circuit Setup

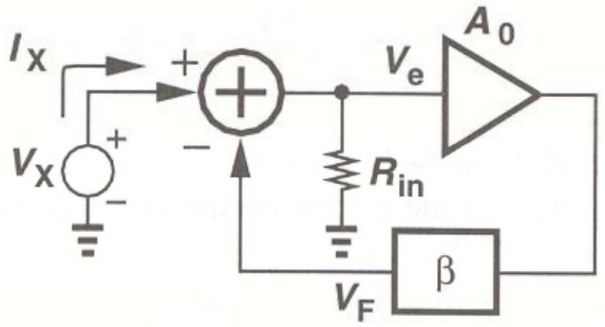
- Small-signal model: Vin drives Rin and the dependent source A₀Ve through Rout to Vout across RL
- Test current Iin injected in series at the input; feedback network β senses Vout in shunt to produce VF, preparing the Rin,closed calculation



# Series-Shunt — Quantitative Closed-Loop Input Resistance

- Voltage sensing ⇒ IF ≈ 0 (ideal-voltmeter assumption at the sensing port), so all of IX flows into Rin
- IXRin(1+A₀β) = VX ⇒ Rin,CL = Rin(1+A₀β) — series-shunt raises Rin and lowers Rout, matching the ideal Voltage Amplifier (Rin=∞, Rout=0)

**voltage sensing => $I_F$ is assumed as 0**




$$V_e = I_X R_{in}$$

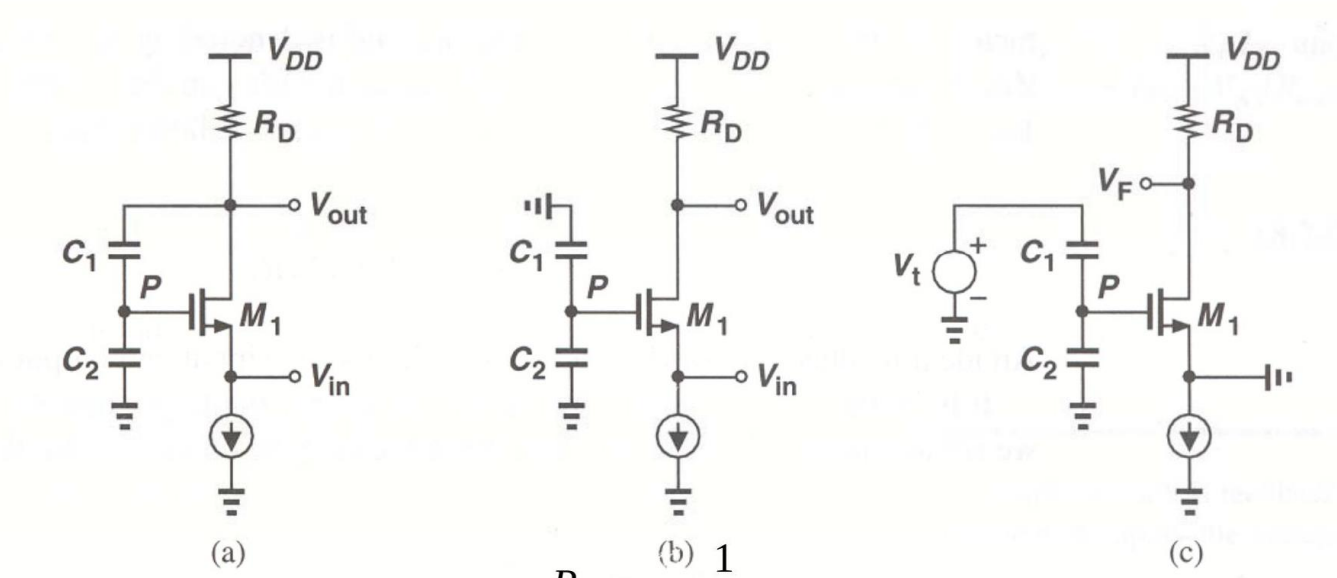
$$I_X R_{in} = V_X - A_o \beta I_X R_{in}$$

$$I_X R_{in}(1 + A_o \beta) = V_X$$

$$\frac{V_X}{I_X} = R_{in}(1 + A_o \beta)$$

# Example 8.3 — Input Resistance of a Common-Gate Stage with Local Feedback

- Common-gate M1 with its gate driven by node P of a C1/C2 divider off Vout, rather than a fixed bias
- Open-loop: Rin,open = 1/(gm1+gmb1); closed-loop: Rin,closed = Rin,open·{1+[C1/(C1+C2)]·gm1RD} — local feedback raises Rin



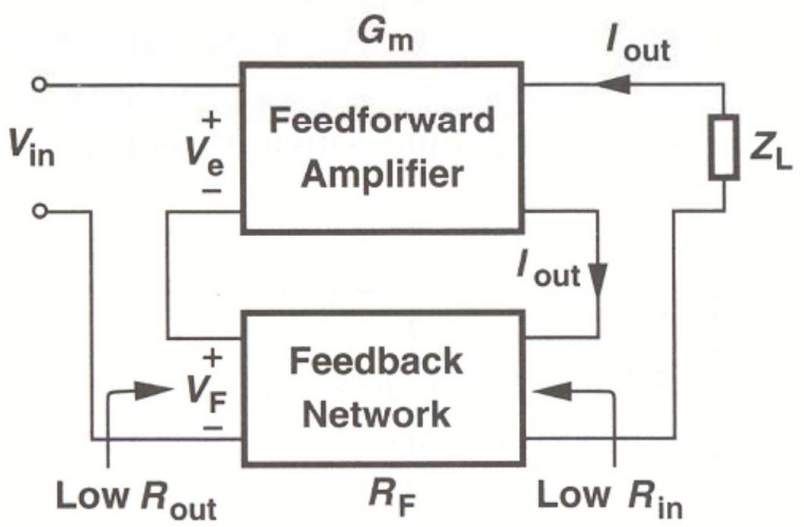

$$V_F / V_t = -\frac{C_1}{C_1 + C_2}(g_{m1} R_D)$$

$$R_{in} = \frac{1}{g_{m1} + g_{mb1}}$$

$$R_{in,closed} = \frac{1}{g_{m1} + g_{mb1}}\left(1 + \frac{C_1}{C_1 + C_2}(g_{m1} R_D)\right)$$

# Series-Series (Current-Voltage) Feedback — Block Diagram

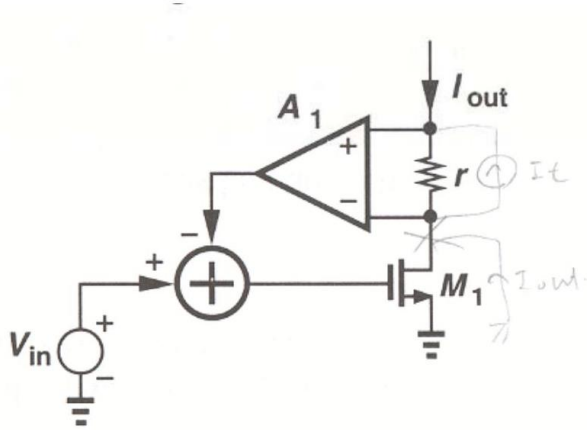
- Feedforward transconductance Gm converts Ve to Iout, sensed in series with load ZL; feedback resistor RF converts Iout to VF, summed in series at the input
- Both feedback-network ports are series connections ⇒ ideally Low Rout and Low Rin at those ports
- Closed-loop gain Iout/Vin = Gm/(1+GmRF)

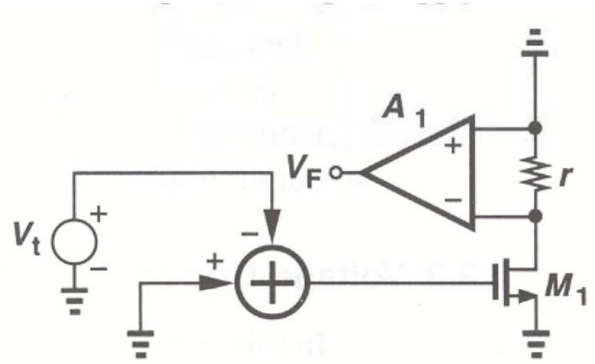



$$\frac{I_{out}}{V_{in}} = \frac{G_m}{1+G_m R_F}$$

# Example 8.4 — Output Resistance with a Sense Resistor and Auxiliary Amplifier

- Common-source M1 with a small series sense resistor r (r≪ro1) at the drain, amplified by auxiliary gain A1 and fed back to the summing node
- Loop gain VF/Vt = −gm1rA1; Rout,closed = (1+gm1rA1)ro1 — current sensing raises Rout, consistent with series-series theory



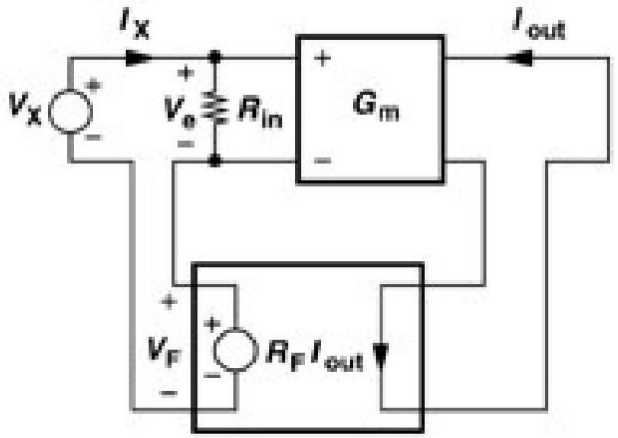

If r is much smaller than $r_{o1}$,

$$V_F / V_t = -g_m r A_1$$

$$R_{out,closed} = (1 + g_m r A_1) r_{o1}$$

# Series-Series — General Closed-Loop Input Resistance

- Test source VX,IX at the input through Rin produces Ve; Gm converts Ve to Iout, and VF=RFIout is subtracted from VX
- Rin,CL = Rin(1+GmRF) — Rin also rises, giving series-series both high Rin and high Rout, matching the ideal Transconductance Amplifier




$$I_X R_{in} G_m = I_{out}$$

$$V_e = I_X R_{in}$$

$$V_e = V_X - G_m R_F I_X R_{in}$$

$$\frac{V_X}{I_X} = R_{in}(1 + G_m R_F)$$

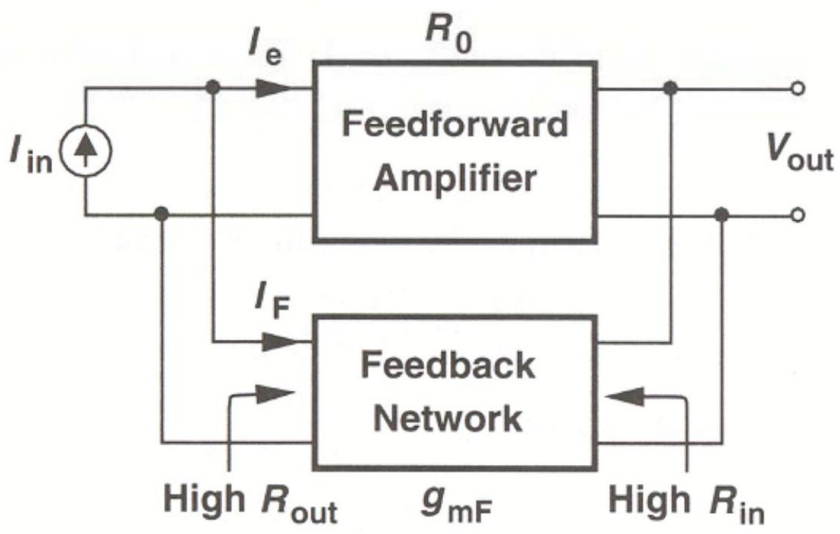
V

# Feedback Topologies II — Voltage-Current & Current-Current

- Shunt-shunt: transimpedance amplifier and the optical-receiver application
- Shunt-series: current amplifier
- Worked examples Ex8.5–Ex8.6

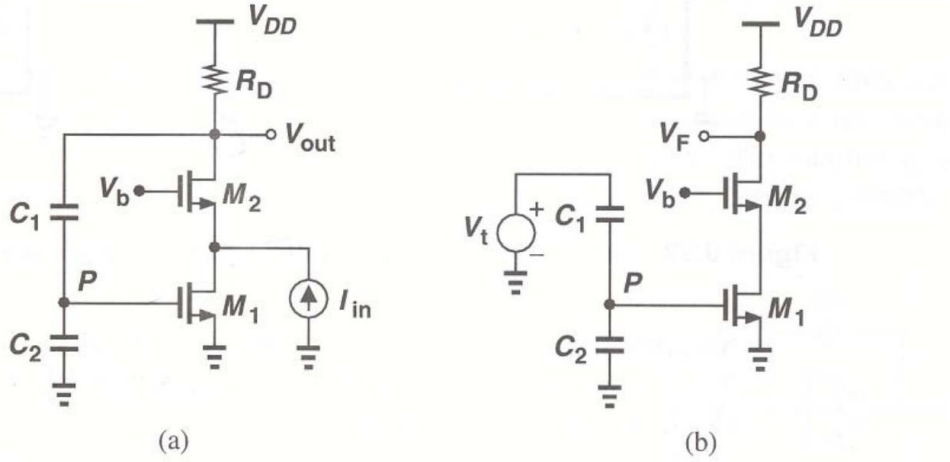
# Shunt-Shunt (Voltage-Current) Feedback — Block Diagram

- Feedforward transimpedance $R_0$ converts error current Ie to Vout; feedback element gmF senses Vout and returns IF, both summed in shunt at the input
- Both feedback-network ports are shunt connections ⇒ ideally High Rout and High Rin at those ports — the opposite pattern from series-series
- Realizes a transimpedance amplifier — a key building block for optical-receiver front ends (see Slide 36)



# Example 8.5 — Transimpedance Gain with Cascode and Capacitive Feedback

- Cascode M2 (fixed Vb) atop input stage M1; Iin injected at the M1/M2 common node; a C1/C2 divider off Vout feeds back into M1's gate at node P
- Loop gain via test injection at P: −[C1/(C1+C2)]gm1RD; Rtot = RD/{1+[C1/(C1+C2)]gm1RD}, reducing the effective transimpedance from the open-loop RD

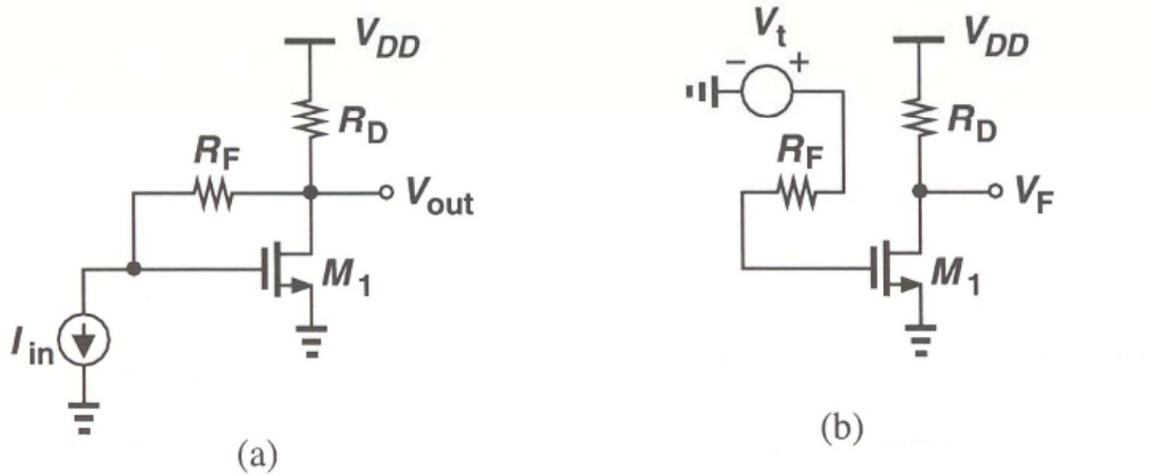



$$-V_t \frac{C_1}{C_1 + C_2} g_{m1} R_D = V_F$$

$$R_{tot} = \frac{R_D}{1 + \frac{C_1}{C_1 + C_2} g_{m1} R_D}$$

# Example 8.6 — Classic Shunt-Shunt Amplifier with Feedback Resistor

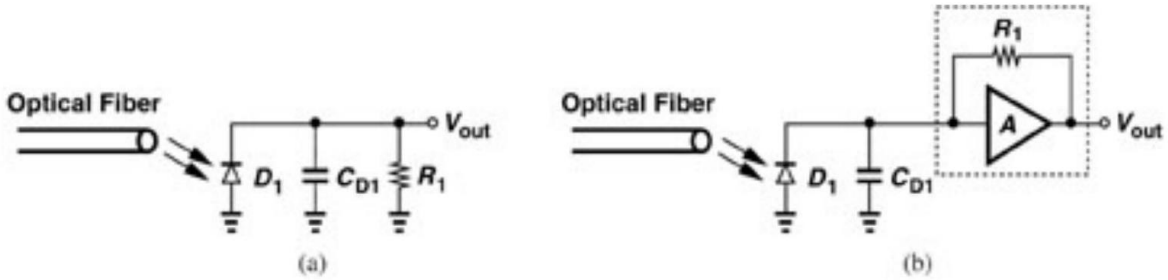
- Common-source M1 with drain resistor RD, and RF connected directly from Vout back to the gate (which also receives Iin) — canonical CMOS transimpedance front-end for photoreceivers
- Rin,closed = RF/(1+gmRD), Rout,closed = RD/(1+gmRD) — both drop with loop gain gmRD, matching the ideal Transimpedance Amplifier (Rin=0, Rout=0)




$$R_{in,closed} = \frac{R_F}{1+g_m R_D}, R_{out,closed} = \frac{R_D}{1+g_m R_D}$$

# Application — Optical-Fiber Receiver Front End

- Photodiode D1 (with parasitic CD1) generates a photocurrent; (a) a passive resistor R1 alone converts it to Vout; (b) an active transimpedance amplifier (Ex8.6 topology) replaces R1
- (b) delivers current into R1 more effectively because feedback gives it a smaller input resistance, diverting less signal current into CD1 at high frequency

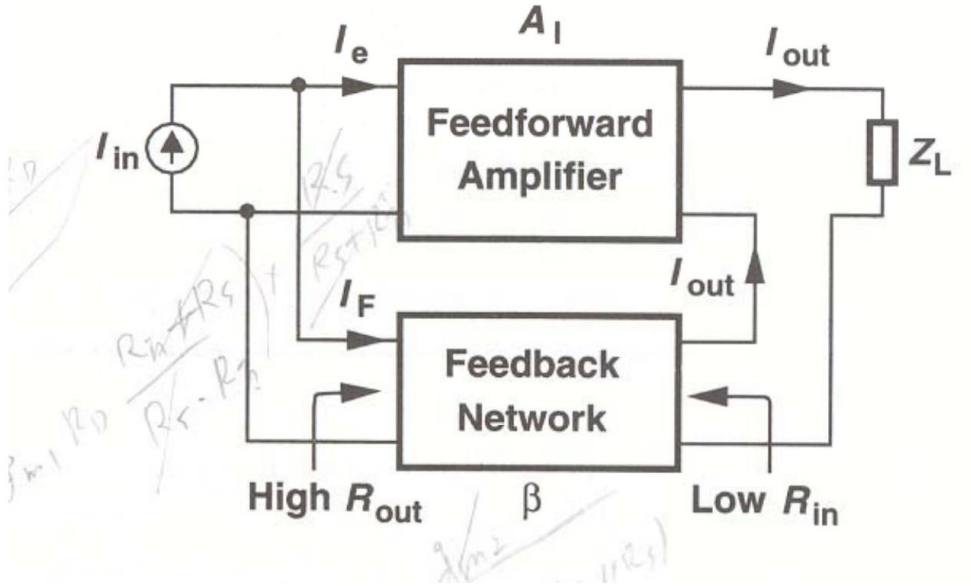



Which one is better in delivering current into the R1?

(b) has a smaller input resistance due to the feedback

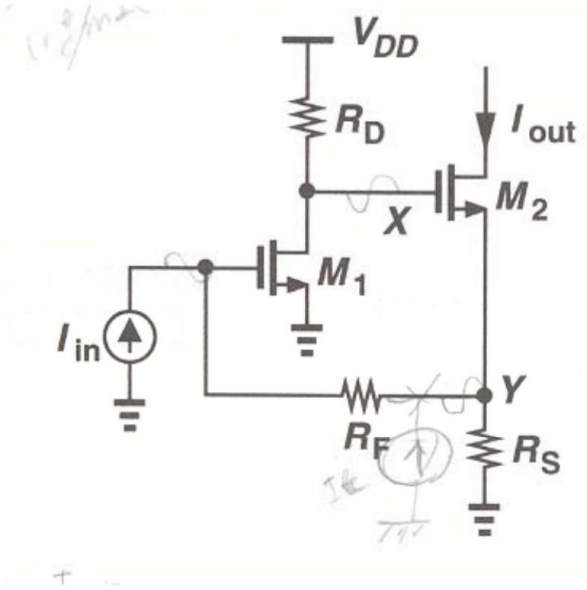
# Shunt-Series (Current-Current) Feedback — Block Diagram

- Feedforward current gain AI converts Ie to Iout, sensed in series with ZL; feedback network returns IF summed in shunt at the input
- Mixed port types on the feedback network: High Rout at the shunt return port, Low Rin at the series sensing port
- Realizes a Current Amplifier (Rin=0, Rout=∞), the dual of series-shunt



# Shunt-Series Circuit Realization

- M1 (common-source, Iin directly into the gate — shunt input) drives cascode-like M2 delivering Iout; RD loads the intermediate node X
- A sense resistor RS in the output path and feedback resistor RF return a current-proportional signal from node Y back to M1's gate — series sensing, shunt return



VI

# Effect of Loading of the Feedback Network

- Two-port network models: Z, Y, H, G parameters
- Computing the loaded open-loop gain
- Worked examples Ex8.7–Ex8.10

# Effect of Loading — Feedback Network Loads the Open-Loop Amplifier

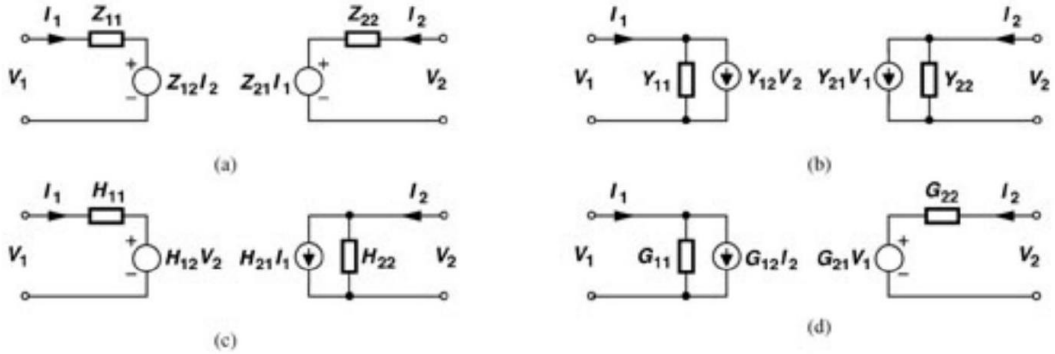
- All earlier derivations assumed an ideal (non-loading) feedback network; real RF, R1/R2 elements have finite impedance and load both amplifier ports
- The remaining slides build a systematic two-port-based method to compute the true loaded open-loop gain Av,open for all four topologies

Feedback network affects the open-loop amplifier

This is an effect of loading of feedback network on the open-loop amplifier

# Two-Port Network Models: Z, Y, H, and G Parameters

- Z-parameters (V1,V2 in terms of I1,I2), Y-parameters (I1,I2 in terms of V1,V2), H-parameters (hybrid: I1,V2 independent), G-parameters (hybrid: V1,I2 independent)
- Model choice depends on whether each feedback-network port is series or shunt; G-parameters naturally fit series-shunt (series output port, shunt input port)




$$V_1 = Z_{11}I_1 + Z_{12}I_2$$
$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

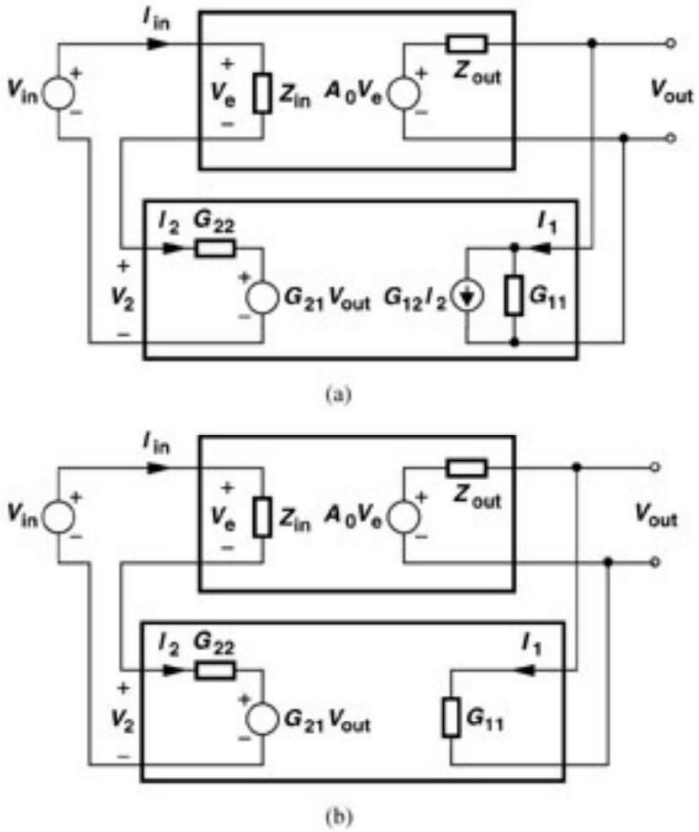
$$V_1 = H_{11}I_1 + H_{12}V_2$$
$$I_2 = H_{21}I_1 + H_{22}V_2$$

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$
$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

$$I_1 = G_{11}V_1 + G_{12}I_2$$
$$V_2 = G_{21}V_1 + G_{22}I_2$$

# G-Parameter Model of the Loaded Series-Shunt Amplifier

- Full model: G22 (input-side self-conductance), G21·Vout (forward feedback current source), G12·I2 (reverse coupling, usually negligible), G11 (output-side self-conductance)
- Simplified G model drops G12·I2, keeping only G22 and G11 as loading elements plus G21·Vout — the standard approximation used throughout the "Effect of Loading" analysis

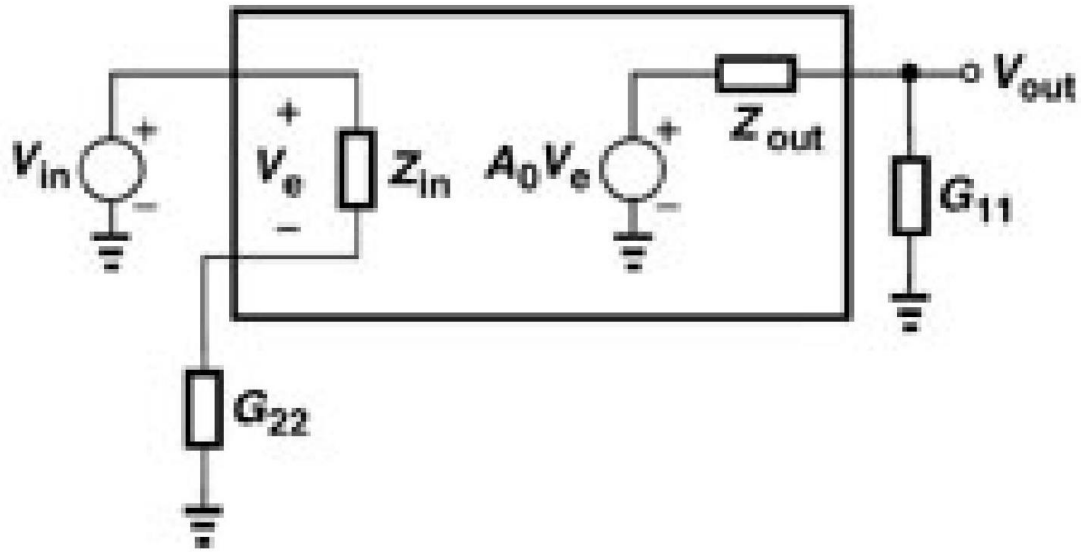



(a) Feedback network represented by G model

(b) Simplified G model

# Reduced Model for Computing the Loaded Open-Loop Gain

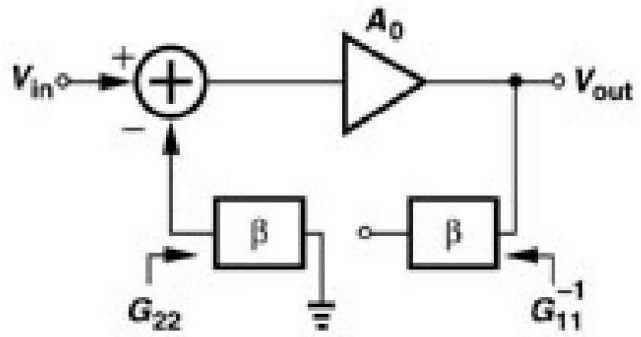
- For Av,open only (not the closed loop), drop the feedback source G21·Vout entirely and keep only the loading conductances G22 (shunt at input) and G11 (shunt at output)
- G22 attenuates Ve via a divider with Zin; G11 attenuates Vout via a divider with Zout — both effects make Av,open smaller than the naively computed unloaded gain




When you see G22, vout should be short to GND
When you see G11, I2 should be zero, i.e, open

# Conceptual View: Opening a Series-Shunt Loop with Loading

- The feedback network β is split into two separate loading elements when the loop is opened: G22 at the input side, $G11^{-1}$ at the output side
- Rule of thumb: keep G22 shunting Vin to ground and treat G11 as an added output-side load — do not simply disconnect β when computing the loaded open-loop gain

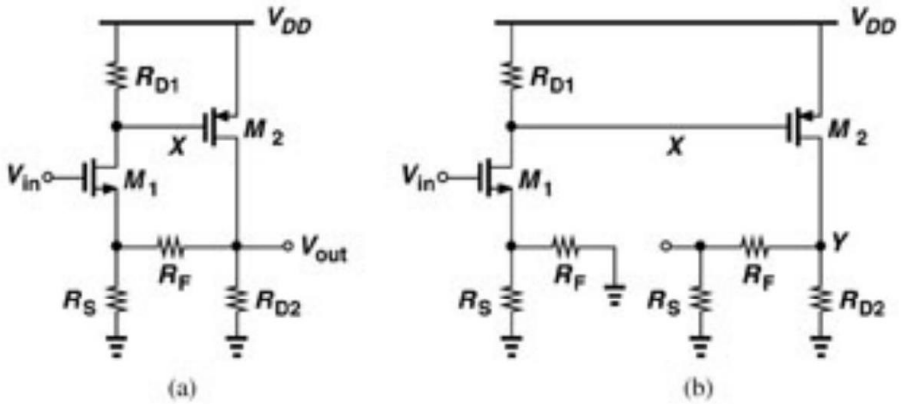



Conceptual view of opening a series-shunt with loading effect

# Example 8.7 — Loaded Open- and Closed-Loop Gain, Series-Shunt

- Two-stage amplifier (M1 CS with RS, M2 cascode to RD1, then RD2 to Vout) with RF feeding back from Vout to M1's source node
- Av,open = [RD1/(RF‖RS+1/gm1)]·[gm2(RD2‖(RF+RS))], β=RS/(RF+RS), T=Av,open·β; Av,closed = Av,open/(1+T)

Ex8.7) calculate the open-loop gain and closed-loop gain



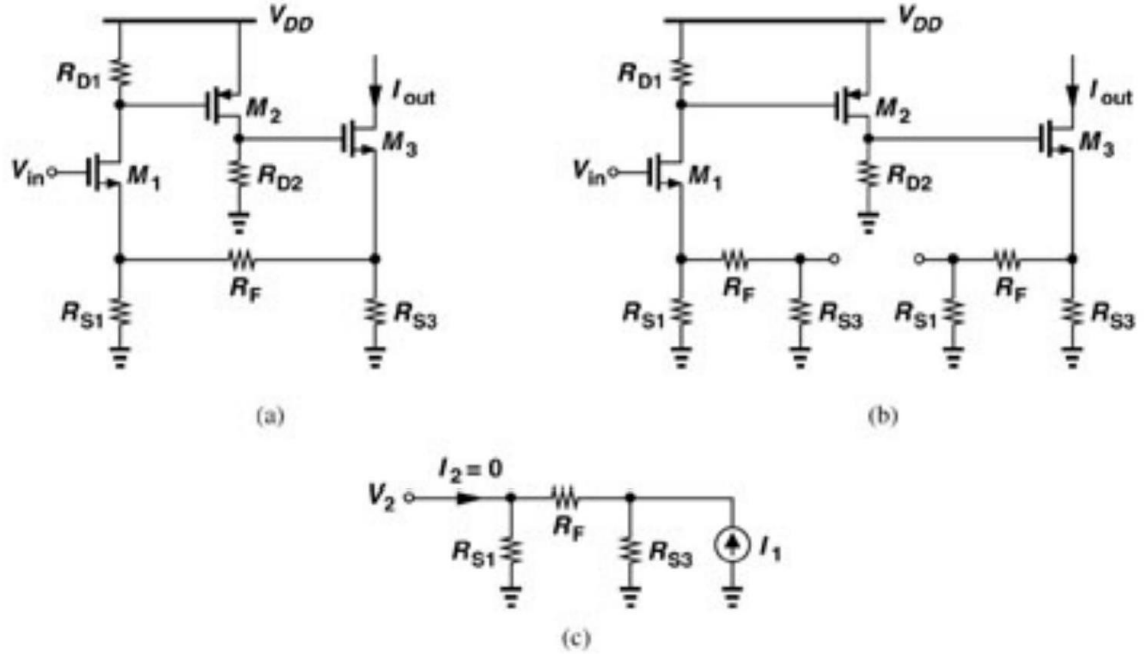

$$A_{v,open} = \frac{R_{D1}}{R_F \,\|\, R_S + 1/g_{m1}}\left[g_{m2}(R_{D2} \,\|\, (R_F + R_S))\right]$$

$$T = A_{v,open}\beta = A_{v,open}\frac{R_S}{R_F + R_S}, A_{v,closed} = \frac{A_{v,open}}{1+T}$$

# Example 8.8 — Loaded Series-Series Amplifier Setup

- Three-stage current-output amplifier (M1 with RS1, cascode M2 with RD1/RD2, output stage M3 with RS3), RF bridging the two source-degeneration nodes
- Loop opened with loading retained on both sides; a Z-parameter model represents the RS1–RF–RS3 feedback network since both of its ports are series connections

Ex8.8) calculate the open-loop gain and closed-loop gain



# Example 8.8 — Loaded Open-Loop Gm and Z21

- Gm,open = {RD1/[RS1∥(RF+RS3)+1/gm1]}·{ gm2RD2/[RS3∥(RF+RS1)+1/gm3] } — three-stage loaded transconductance
- Z21 = [RS3/(RS3+RS1+RF)]·RS1 — the feedback network's reverse Z-parameter, playing the role of β in Gm,closed = Gm,open/(1+Gm,openZ21)

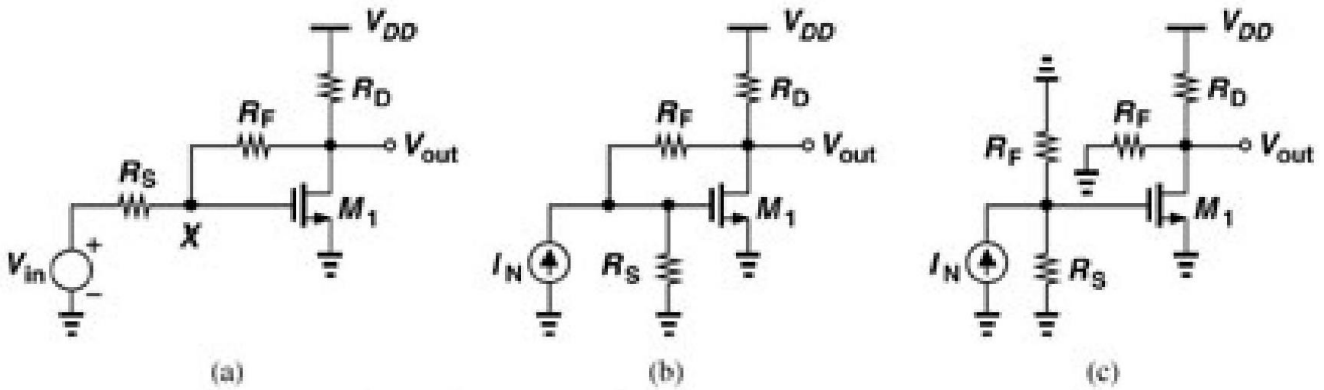
$$G_{m,open} = \frac{R_{D1}}{R_{S1} \| (R_F + R_{S3}) + 1/g_{m1}} \frac{g_{m2} R_{D2}}{R_{S3} \| (R_F + R_{S1}) + 1/g_{m3}}$$

$$Z_{21} = \frac{R_{S3}}{R_{S3} + R_{S1} + R_F} R_{S1}$$

# Example 8.9 — Loaded Shunt-Shunt Amplifier

- Common-source M1 with RF from Vout back to the gate; Ropen = (RS‖RF)·gm·(RF‖RD) accounts for RF loading both the input source and the RD load
- Y21 = 1/RF (Y-parameter feedback coefficient); full closed-loop gain Vout/Vin = (1/RS)Ropen/{1+gm(RF‖RD)[RS/(RS+RF)]}

Ex8.9)




$$R_{open} = (R_S \parallel R_F) g_m (R_F \parallel R_D)$$

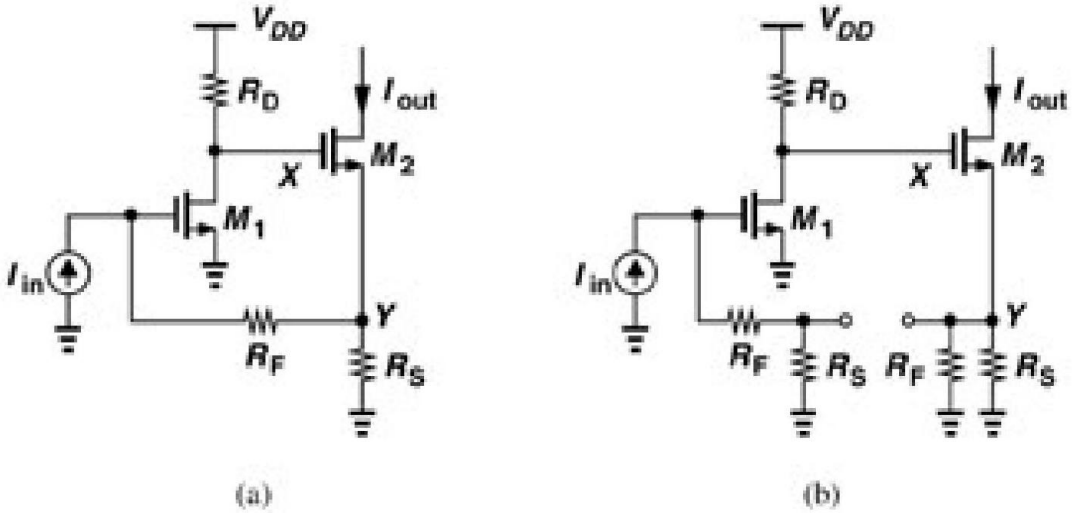
$$Y_{21} = 1/R_F$$

$$V_{out}/V_{in} = \frac{1}{R_S} \frac{(R_S \parallel R_F) g_m (R_F \parallel R_D)}{1 + g_m (R_F \parallel R_D) \dfrac{R_S}{R_S + R_F}}$$

# Example 8.10 — Loaded Shunt-Series Amplifier (Chapter Closing Example)

- M1/M2/RF/RS circuit from Slide 38, now solved quantitatively: AI,open = (RF+RS)gm1RD/[RS∥RF+1/gm2]
- H21 = RS/(RF+RS) (H-parameter feedback coefficient) — closes the chapter's four-topology, loaded-gain treatment using Z/Y/H/G two-port models

Ex8.10)




$$A_{I,open} = (R_F + R_S) g_{m1} R_D \frac{1}{R_S \parallel R_F + 1/g_{m2}}$$

$$H_{21} = \frac{R_S}{R_F + R_S}$$